Global Economics Case Competition (GECC 2024)

The Dichotomy of Renewable Energy

Team **The Philosopher's Stone**

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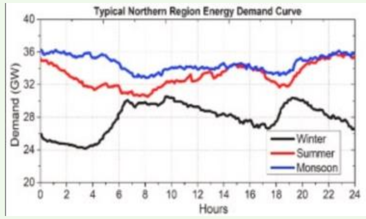
Current Energy Landscape

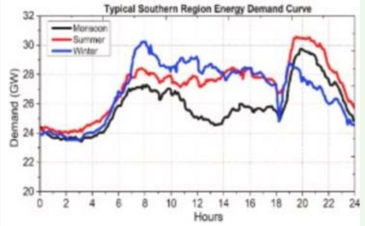
PROBLEM STATEMENT

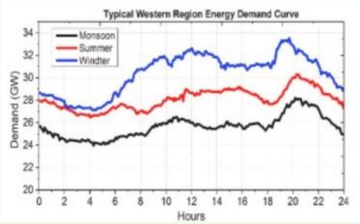
Amidst global endeavors to address climate change and shift towards sustainable energy, the Indian government grapples with the intricate task of managing the paradox of renewable energy, balancing ambitious targets with pressing issues for a smooth transition.

Energy Needs & Sources

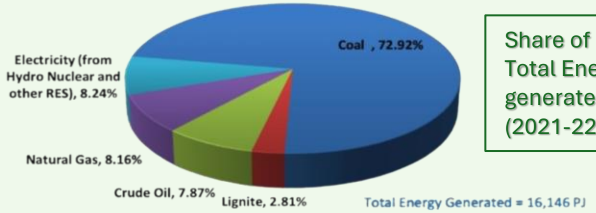
- India, the world's most populous country, **ranks third in both energy production and consumption**, with 1380 TWh/year and 1190 GWh/year from all sources (IEA, 2019).
- In fiscal year 2018–2019, India generated 1551 TWh of power, with renewable energy accounting for 4% (62 TWh). Per capita consumption increased from 1149 kWh to 1181 kWh per year, with an average growth rate of 3%
- Peak electricity consumption in India has improved over time, with a 0.6% deficit in 2019 compared to the required 1274 BU, suggesting increased electrification, notably after 2009.
- The western zone saw a rise in power demand to 388 BU, constituting around 30% of total demand, with the northern and western zones combined accounting for over 60% of India's electricity consumption.
- **The dominant energy consumption zones** in India are the **northern, western, and southern regions**, highlighting the need to track regional electricity demand changes across seasons for balanced energy distribution and predicting load shedding.
- **Hourly electricity demand varies seasonally**: the north peaks during the monsoon and dips in winter, the west peaks in winter and dips in the monsoon, while the south peaks in the monsoon and drops in winter, aiding grid management and load forecasting.







India still depends heavily on coal as the major source of energy. During the FY 2021-22, energy generated from Coal accounted for about 72.92% of the total generation of energy followed by Electricity (from Hydro, Nuclear and other Renewable energy sources) (8.24%) and Natural gas(8.16%).



Share of Total Energy generated (2021-22)

Existing Infrastructure

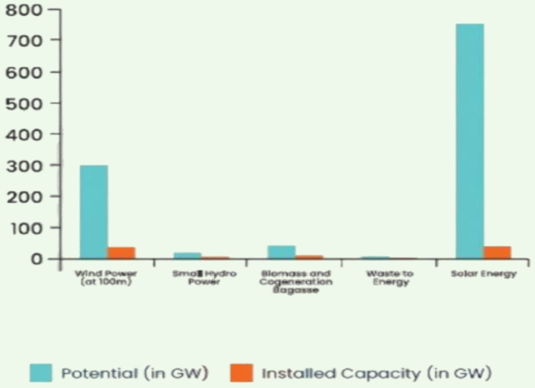
Potential and Installed Capacity of Renewable Energy Sources

- India's renewable energy potential, as per **MNRE's 2019 report**, is estimated at 1100 GWp, encompassing various sources like wind, hydro, biomass, and solar.
- Despite significant growth in renewable energy capacity over the past two decades, MNRE's 2022 data indicate that current capacity remains below one-tenth of the total potential as of December 2021.
- **Solar and wind power dominate the renewable energy sector** due to their extensive potential, installed capacity, and rapid growth.
- Although renewable energy capacity has expanded, most India's electricity still comes from conventional sources, with thermal sources comprising 60% of installed capacity but only contributing 73% of total electricity generation.

Potential and Installed Capacity of Thermal Energy Sources

- **Coal, oil, and gas usage in India constitutes almost 85%** of total energy consumption, surpassing proven reserves twofold, with the Ministry of Petroleum and Natural Gas reporting 493.26 million metric tons of oil and 1080.42 billion cubic meters of natural gas in 2021.
- Coal contributes over 70% to India's electricity consumption, totaling 1342 TWh, and maintains a staggering 95% dominance in the thermal electricity mix.
- Despite possessing a thermal installed electrical capacity of 235 GW, equating to 60% of the total installed capacity, coal's actual consumption share stands at 75%.
- Apart from its role in electricity generation, coal finds extensive use in the brick, cement, and steel industries.

Renewable Energy Potential vs Capacity



Environmental Concerns

In 2020, India had 22 cities among the world's top 30 most polluted according to PM2.5 levels.

India is the world's third-largest GHG emitter responsible for 6.6% of global emissions in 2020.

According to a World Bank report, water pollution costs India about 3% of its GDP annually.

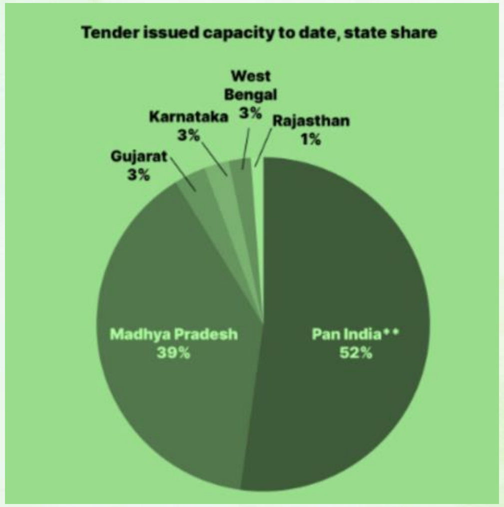
Renewable Energy Reliability Innovations

Energy Storage

Virtual Power Plant
A VPP aggregates diverse DER capacities like solar, batteries, and EVs, enabling power generation, trade, and market participation, akin to internet-enabled desktops versus mainframes.

Pumped Hydroelectric storage
Pumped hydroelectric storage pumps water uphill during low-demand, low-price periods, releasing it downhill through turbines during high-demand times to generate power.

Gravity Based System
Energy Vault's gravity-based system elevates 30-tonne bricks using renewable energy, storing potential energy, then releases it as kinetic energy into power grids.



Grid Integration

Reimagining operation and planning for a reliable, cost-effective, and efficient electricity system with cleaner new energy generators.

Modeling and Optimization

Developing capabilities in modeling, control, and optimization for renewable-dominated power grids, utilizing power electronics and data analytics.

Resilience Enhancement

Advancing prediction capabilities to support network evolution to withstand extreme events like wildfires, tsunamis, hurricanes, or cyber-attacks.

Technology Integration and Optimization

Optimizing interconnected technologies (e.g., generators, electric loads, and storage) at various scales to enhance operations, efficiency, and cost-effectiveness, while reducing reliance on peaking or emitting facilities.

Network Analysis and Planning

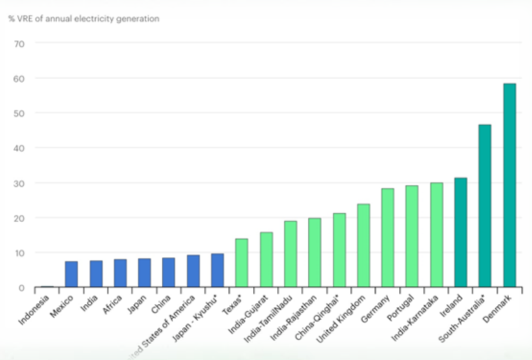
Analyzing meter-, microgrid-, feeder-, substation-, and community-scale networks for planning and optimization purposes, including the development of new economic frameworks that encompass a full range of renewable energy services and costs.

As of December 2023, India's renewable energy sources have a combined installed capacity of 180.79 GW, including 73.31 GW of solar power and 44.73 GW of wind power

The Government of India plans to increase renewable electricity capacity to 175 GW by 2022 and to 450 GW by 2030.

Hybrid Systems

- **The IEA system** integration framework for renewables is divided into six phases, each of which provides guidance on how to efficiently manage integration difficulties
- Most countries worldwide are in Phases 1 and 2, with minor integration issues, whereas India's renewables-rich states already confront considerable challenges
- These obstacles are caused by causes such as rising hourly demand unpredictability, increased ramping requirements because of solar energy's influence, and issues with short-term frequency swings and local voltage
- India's states with abundant renewable resources have a larger share of **variable renewable energy (VRE),** which exacerbates system integration challenges compared to many other countries
- Recognizing and overcoming these problems is critical to ensure the seamless integration and dependable operation of renewable energy sources in India's power grid



In March 2020, GSECL completed a successful experimental run at its 500MW Ukai Thermal Power Plant, operating at a minimal load of 40% while maintaining stability, with the goal of assessing the flexibility of integrating 30GW of variable renewable energy (VRE) onto Gujarat's grid. Similar phased VRE integration can be emulated in states with substantial renewable energy generation, improving grid flexibility and reliability in the medium run. In addition, states with existing wind and wave energy infrastructure, such as Karnataka and Kerala, can combine renewable energy sources that compliment one another.4. In a similar way, different renewable energy sources can be integrated

Forecasting and Predictive Analytics

Data science optimizes renewable energy using advanced algorithms and analysis.

Predictive analytics enhances data science by forecasting future trends, aiding in predicting energy production, and consumption patterns

Machine learning improves the accuracy of predictions over time, while addressing challenges such as data security and ethical considerations

Economic opportunities and Job Creation

As India grapples with the challenges of climate change, energy transition emerges not only as a solution to environmental concerns but also as a potent driver of economic growth and job creation.

How?

The transition to renewable energy will stimulate economic growth by creating jobs, providing investment possibilities, to name a few

Then?

The transition will lead to lowering costs, promoting energy independence, innovating, providing environmental advantages, promoting rural development

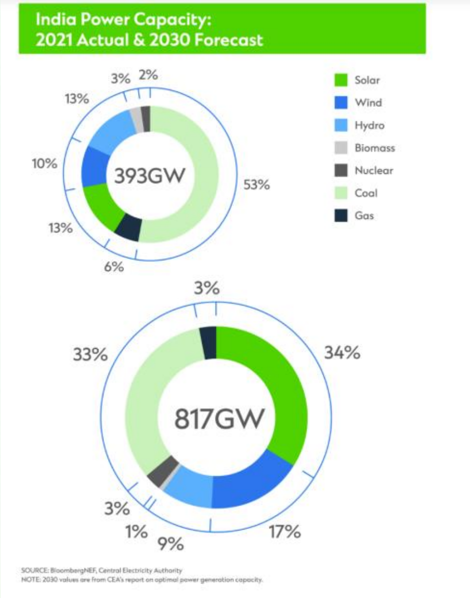
Result

It will foster economic growth coupled with sustainable developmental goals for a better tomorrow

Economic Opportunities

India's energy mix has traditionally been dominated by coal majorly. The transformation to green energy is changing the dynamics of energy mix in India. India's power infrastructure is mostly dominated by coal, which accounts for more than 50% of capacity. However, the share of fossil fuel-based energy is 70 to 74 per cent.

Decoupling India's economic growth from greenhouse gas emissions is a massive, urgent project. Indian conglomerates are already tuned into this and leading the way by mobilizing fast.
Earlier last year, Adani Group announced a strategic partnership with Total Energies to jointly create the world's largest green hydrogen ecosystem with an ambitious investment target of USD50 billion over the next 10 years in green hydrogen and associated ecosystem.
In the first phase, the joint venture is looking to develop green hydrogen production capacity of 1 million ton per annum before 2030.



Job Creation

- The transition to a carbon-neutral economy in India would close coal mines and thermal power plants, affecting people throughout the coal value chain
- Blue-collar workers with limited education and skills are vulnerable due to their communities' reliance on coal-related income
- Central and local governments must plan for the transition by providing tailored solutions, ongoing dialogue, and financial support
- Effective planning necessitates the mobilization of finances, policy measures, and continuing engagement to reduce disruption and help affected communities
- Indirect workers and local companies related to the coal sector also need specialized aid to handle economic obstacles throughout the transition

Skill Development

- Skill and education programs
- Employment and job matching

Social Development

- Temporary financial support
- Employment Services

Worker Preferences Alignment

Young Workers	Low Preference	Re-skilling
Mature Workers	High Responsibility	Entrepreneurship
Elderly Workers	Close to retirement	Social benefits

Unlocking opportunities by the GOI

- MSME promotion
- Job Creation by PSUs
- MGNREGA and other programs

IIM SHILLONG

Land Use and Environmental Impact

Preface

- Green India Mission prioritizes forests' importance in climate mitigation, biodiversity preservation, and community livelihoods.
- GIM aims to enhance forest cover and quality across 10 million hectares over a decade, aligning with broader environmental and socioeconomic objectives.
- With India's population surpassing that of China, the demand for land intensifies, necessitating housing, food, and energy solutions.
- Proactive land use planning is crucial to align renewable energy expansion with forest conservation, agricultural needs, and urban development, reducing conflicts and promoting sustainability.
- Utilizing degraded lands for renewable energy development presents a practical approach to minimize environmental harm while effectively achieving energy goals, without encroaching on valuable natural habitats or agricultural lands.

Renewable Energy and Mitigation Integration

Balancing renewable energy development with forest conservation and social goals requires careful planning to minimize impacts and avoid conflicts over land use

- Regulatory Overside through EIA
- Incentivize low impact development practices
- Develop compliance standards for utilities
- Protect productive agricultural lands
- Balance renewable energy expansion
- Promote offshore wind, rooftop solar, wave energy etc.
- Assess land with standards to avoid regional concentration
- Focus on Agrivoltaics sector to secure benefits to farmers

Affordability and Financial barriers

Financial Barriers

- Reluctance of banks to lend for RE projects
- Regulatory concerns delaying government loans
- Institutional and policy complexities hinder progress
- Short tenures and high debt costs
- Information asymmetry in banking sector
- Risks due to technological leapfrogging

Agencies financing Renewable Energy projects

इरेडा IREDA

पी एफ सी PFC

आर ई सी REC

Initiatives

Recognition of Priority Sector Lending by RBI.

Infrastructure Debt Fund

Soft Loans from IREDA.

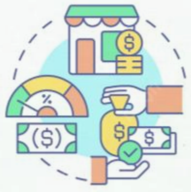
Feed-in-tariff (FiT) schemes

National Clean Energy and Environment Fund (NCEEF)

Renewable energy certificates (REC)

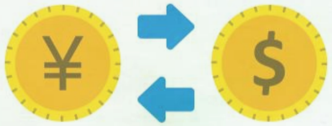
Renewable portfolio obligations (RPO)

Proposed Solutions



Loan Guarantee Program

Underwrite loan repayment



Government to assume currency risks

Facilitate foreign investments

Socialize New Transmission cost

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Thank You